

COSMOTRONIC

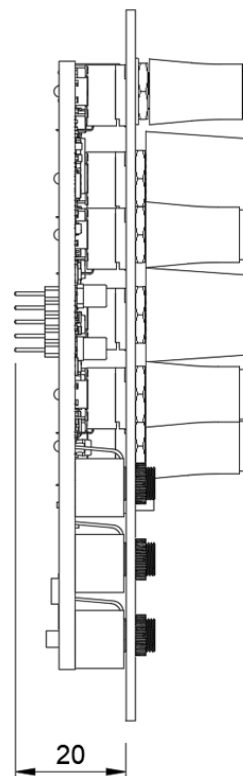
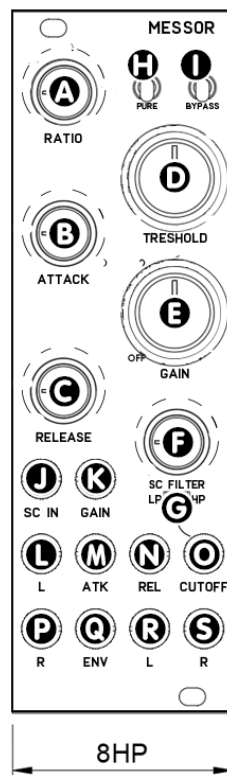
8 HP
REVERSE POLARITY
PROTECTED
+12V: 45mA
-12V: 26mA



#00004
MESSOR

STEREO COMPRESSOR

- A** RATIO
- B** ATTACK TIME
- C** RELEASE TIME
- D** THRESHOLD
- E** MAKE UP GAIN
- F** SIDE-CHAIN FILTER CUTOFF
- G** LOWPASS/HIGHPASS SELECT
- H** WARM SATURATION SWITCH
- I** BYPASS SWITCH
- J** EXTERNAL SIDE-CHAIN IN
- K** GAIN CV IN
- L** LEFT IN
- M** ATTACK CV IN
- N** RELEASE CV IN
- O** SC FILTER CUTOFF CV IN
- P** RIGHT IN
- Q** SC ENVELOPE OUTPUT
- R** LEFT OUT
- S** RIGHT OUT



OVERVIEW

Messor is a stereo compressor with lots of tricks up its sleeve. From glueing a mix together to squashing drums and sidechaining kicks, this versatile dynamics manager does it all.

At its core is a VCA based feed-forward compressor with a low noise signal path. The magic is in the side-chain that shapes the gain reduction. This starts with a precision envelope follower, and is then shaped by control parameters familiar to a compressor: the attack, release, threshold, ratio and makeup gain.

The ratio can go past limiting and into over-compression, which can completely reshape transients, a great tool for sound design. A very useful extra feature is the sidechain filter, switchable between high-pass and low-pass to really dial in the compression response.

The gain reduction envelope is also available as an output, so you can patch it into filters, vca's or even pitch inputs for many creative explorations using compression as a starting point.

Being modular, of course there are CV inputs for Attack, Release and the side-chain Filter cutoff. The Gain CV controls the makeup gain, but also doubles as a Stereo VCA with the Gain knob turned fully CCW.

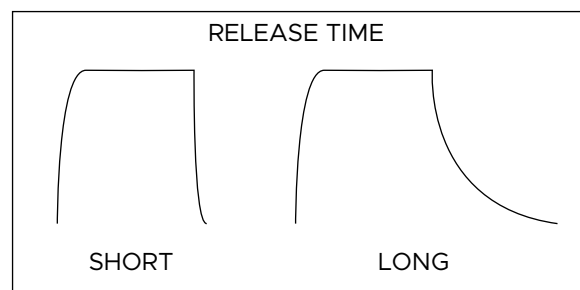
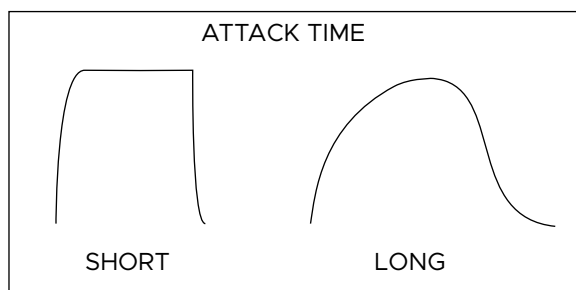
There is a bypass switch, and a Warm mode, that can bring in a bit of additional saturation and compression in the form of analog tube simulation circuitry. Lastly there's the possibility to insert an external sidechain, for that classic kick sidechain compression.

Attack

The attack knob ③ sets the time the compressor takes to start reducing the output volume. In other words, this setting controls how much of the initial transient is left intact, and how fast the compression kicks in.

Longer settings average out the overall dynamics, giving a much smoother envelope, and more subtle compression. The attack behaves similar to a low-pass filter, so also it affects how the compressor handles high frequencies. Longer attack times also extend the release.

The attack time can be modulated with CV, the input jack for this is labelled ATK. The useful input range is 0-5v, with the Attack knob working as an offset.



Release

The Release knob ④ sets the Release time of the Gain Reduction envelope, from around 5ms to 3,5s. The Release means how long it takes for the Compressor to stop reducing the gain, after the input signal is below the Threshold again. Similar to the Attack, longer Release times create an averaging effect.

Very short Release times can create a distortion in low frequencies, as the compression cuts off part of the waveform. There is a CV input labelled REL, with a useful input range of 0-5v, with the Release knob working as an offset. Negative CV can push the Release time below 5ms.

Threshold

This knob ⑤ sets the level where the compression kicks in. All signal levels under the Threshold are ignored, while all signal levels above it generate the side-chain. This side-chain is then further shaped by the attack and release settings to generate the gain-reduction envelope. This envelope is visualised by the orange leds under the Threshold knob, to give you a handy indication of what is going on under the hood.

Setting the Threshold lower will create a more gain reduction. Please note that this module expects Modular levels (-10 to 17dBv) so it won't work with line level signals.

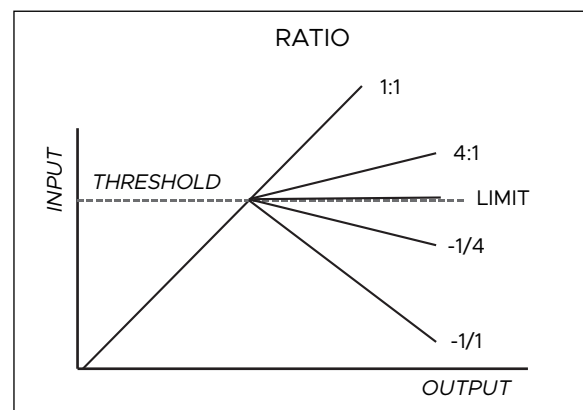
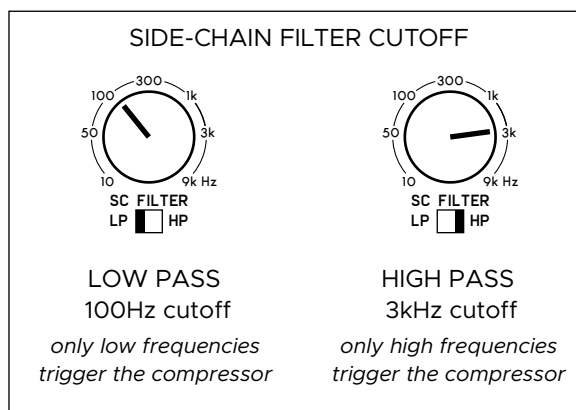
Side-chain Filter

The side-chain is filtered with a 6dB switchable low-pass / high-pass filter ④ . This can have a dramatic effect on the sound, as certain frequencies trigger gain reduction while others leave the signal unchanged.

For example, switching the filter to High-Pass (HP) ⑤ and setting the cutoff to 300 Hz filters out low frequencies like a kick drum, that might otherwise overpower the compression.

One thing to be aware of is that if the filter can be set to cut off all frequencies, for example a Low Pass with a cutoff at 30Hz. Now there won't be any compression, since there is no side-chain signal coming through!

The cutoff of the filter can be modulated using the CUTOFF CV input.



Ratio

Turning the Ratio ⑥ all the way counter clockwise turns off the compressor. Slowly turning up the Ratio knob starts bringing in compression, roughly following the ratios indicated on the knob. For subtle compression it's best to keep it in the lower ratios, up until the 12 o'clock position.

In this position Messor functions more as a limiter, restricting signals to the Threshold level (with GAIN at 0).

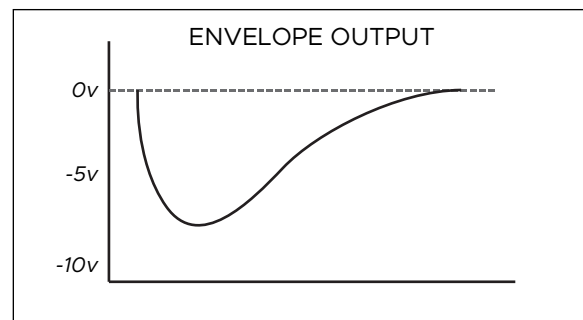
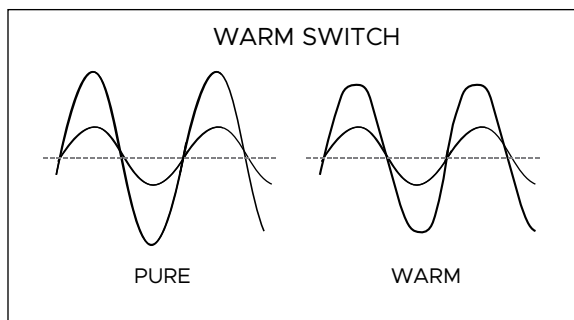
Turning the knob further pushes past limiting into negative ratios. This means an input signal that goes over the threshold has its output volume reduced to below the threshold. With the Ratio knob fully clockwise, and a low Threshold, loud signals become very quiet. This can be used creatively in combination with extra Make Up gain, to create a dynamic expansion.

Please keep in mind that the actual gain reduction depends on a combination of all settings. Input levels, threshold, attack, release and filter settings all affect each other, so the labels on the knob are therefore more of a general indicator than an exact setting.

Make Up Gain

The make up GAIN ③ is the gain applied to the signal after the compression takes place. This lets you match the output signal to the input signal, by bringing up the overall level. Different settings on Messor might require different Make up Gain levels to compensate for the gain reduction.

There is a CV input labelled GAIN that can also modulate the output level. The MAKE UP knob acts as an offset in this case. You can use this Messor as a stereo VCA as well, set the Knob to MUTE and patch an envelope to the CV input. Nominal CV values are 0-5v. Using CV can also give you even more gain.



Warm / Pure Switch

In some settings with lots of gain and longer attacks you might find the first part of the transient starts clipping. This happens because the loudest part is amplified more, while the gain-reduction hasn't kicked in yet.

In these scenarios the *warm* ④ mode comes in handy, a tube saturation emulator. Enabling this circuit will increasingly soft clip the signal, as the headroom limit comes closer, while leaving the lower level signals intact. This adds some extra compression and gives a warmer, more saturated sound. The *pure* mode disables this effect.

A small side effect of the warm mode is that output volume is a little lower, this can be compensated by adding some extra make-up gain if needed.

ENV Output

The gain reduction envelope is also available as an output ⑤. It's a negative envelope, since its goal is to reduce a value.

If you have an oscilloscope it can be very handy to visualise this envelope, as it allows you to precisely dial in the shape of the gain-reduction.

The ENV Output is available independent of Ratio and Bypass switch settings.

Left and Right input

The Left and Right **1** **2** are true stereo inputs.

The Left input is normalised to the Right input, if nothing is patched into it.

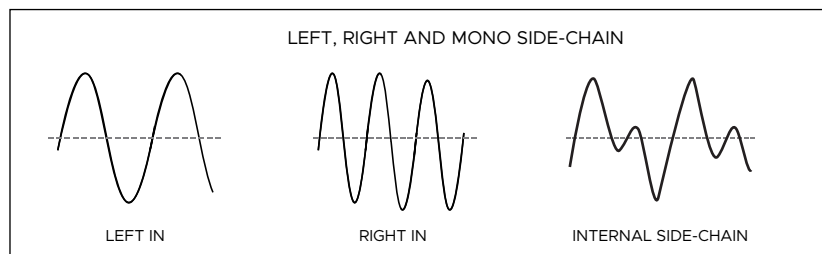
Input impedance is 30k.

External Side-chain input

This CV input **3** is normalised to a mono mix of the left and right channel.

Patching into this input breaks the connection, and allows other signals to affect the side-chain.

Patch in a kick drum for some classic pumping side-chain compression.



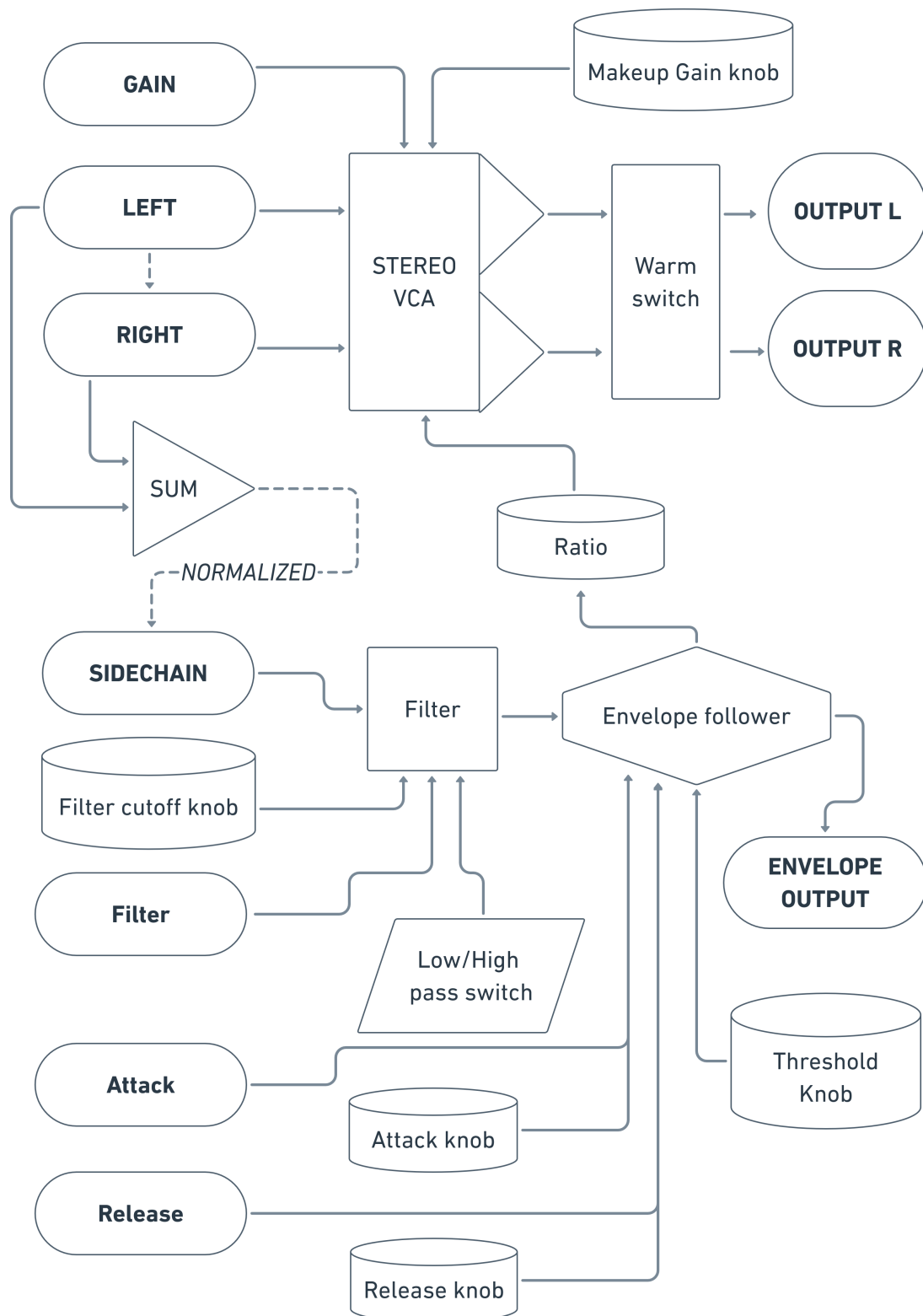
Getting started

There's a great variety of sounds and sweet spots to explore, so please experiment and use your ears.

To get started I'd recommend the settings shown here.



MESSOR SIGNAL FLOW



POWER

MAKE SURE THE POWER IS OFF WHEN INSTALLING A NEW MODULE IN YOUR SYSTEM, AND PAY CLOSE ATTENTION TO THE ORIENTATION OF THE POWER CABLE.

PLEASE NOTE THE LABEL "RED" AND WHITE LINE, UNDER THE POWER HEADER ON THE BACK OF THE MODULE. MATCH IT WITH THE RED LINE OF THE RIBBON POWER CABLE. THE RED LINE INDICATES THE $-12V$ POWER RAIL.

COSMOTRONIC MODULES ARE REVERSE POLARITY PROTECTED, BUT WILL NOT TURN ON IF THE CABLE IS REVERSED.

LIMITED WARRANTY

COSMOTRONIC WARRANTS THIS PRODUCT TO BE FREE FROM DEFECTS IN WORKMANSHIP AND MATERIALS FOR A PERIOD OF ONE YEAR FOR THE ORIGINAL INVOICE DATE. THE WARRANTY COVERS REPLACEMENT OR REPAIR.

FINAL DISCRETION OF WARRANTY COVERAGE LIES SOLELY WITH COSMOTRONIC.

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